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Self supporting solar solution

Abstract

An electrical bus system has a trunk bus that includes at least one positive bus line, an equal number of negative bus line, and a ground cable. The trunk bus lines extend through clamping blocks where some of the clamping blocks are coupled to brackets that are themselves coupled to vertical supports. The ground cable is coupled to vertical support using the bracket. Multi-tap shear bolt connectors are used to connect the trunk bus line to photovoltaic wires carrying the output from one or more solar panel arrays, thereby eliminating the need for combiner boxes used in conventional systems.

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Background/Summary

PRIOR APPLICATION (1) The present application is a continuation of U.S. Utility patent application Ser. No. 18/989,980 filed Dec. 20, 2024, which claims priority to and benefit of U.S. Provisional Patent Application No. 63/613,885 filed Dec. 22, 2023 and U.S. Provisional Patent Application No. 63/680,776 filed Aug. 8, 2024, all of which are hereby expressly incorporated by reference.

TECHNICAL FIELD AND BACKGROUND

(1) The present invention relates generally to electrical bus systems, and more particularly, to an electrical bus system that interconnects arrays of solar power generation units.

(2) Conventional solar power generation systems feed the output current from numerous solar panel arrays to combiner boxes. Electrical buses carry the current from combiner boxes to a centralized power system and in some cases, carry current from the solar panel arrays to the combiner boxes. There is a desire in the industry to lower cost and increase efficiency by using less components, such as fewer photovoltaic wires and connectors, which in turn requires the components that are used to carry higher current amperage.

(3) One disadvantage of increasing current is that combiner boxes are prone to overheating and failure even from relatively small losses at the numerous connection points and the accompanying high currents, and combiner boxes are difficult to safely service and replace. Conventional electrical bus lines connecting the combiner boxes also suffer from disadvantages that include high material costs and significant labor required to install the bus systems. Electrical bus systems are generally buried in trenches, which require heavy machinery to dig and prepare the ground to

receive the bus systems, or they are hung with hangers suspended above ground together with components that must be coupled to the electrical bus.

(4) Given the drawbacks of conventional systems for electrically interconnecting solar panel arrays, it is an objective of the present invention to provide electrical bus systems that eliminate the need for combiner boxes and that are less expensive and more efficient to install. The bus disclosed in this application includes a self-supporting solution that provides mechanical support and that is used to couple the bus to vertical supports for installation without the need for hangers that are labor intensive to install and maintain. The bus is coupled to vertical supports using brackets that can be installed without hardware.

(5) The individual trunk bus lines run through separate cable throughways formed in clamping blocks that separate the trunk bus lines to promote cooling. The clamping blocks attach to the brackets that are coupled to the vertical supports.

(6) The positive and negative bus lines can be formed from a ruggedized cable construction suitable for both low voltage and high voltage applications. The ruggedized cables can include an aluminum coated inner steel member to add rigidity and strength to the bus lines. The bus line is coupled to string photovoltaic wires coming from the solar panel arrays using a multi-tap shear bolt connector that eliminate the need for combiner boxes and that show favorable temperature performance compared to combiner boxes.

SUMMARY

(7) In one embodiment of the electrical bus system, a plurality of vertical supports are arranged in a row where each vertical support is proximal to at least one power generation unit, such as one or more solar panel arrays. A trunk bus extends between the vertical supports. The trunk bus has a positive bus line, a negative bus line, and an aluminum coated steel core member running through the interior of the positive bus line and the negative bus line. A bracket is fixed to each of the vertical supports without the use of hardware. The trunk bus is secured to the bracket at each vertical support by running the trunk bus lines through a clamping block coupled to the bracket.

(8) In one embodiment, the bracket has a head, a neck, a flattened body, and one or more openings. The head and the neck extend through a slot in the vertical support and the clamping block is secured to the body of the bracket. In another embodiment, the bracket is an "L-shape" with a body, a free end and a coupling end, a rabbeted portion extending from the coupling end, and a stem extending from the rabbeted portion. The stem extends through the slot in the vertical support. In another embodiment, the bracket is a "C-shape" with a vertical coupling portion, a first end, a second end, and a front and rear surface. A first portion extends perpendicular from the front surface of the vertical coupling portion first end. A second portion extends perpendicular from the front surface of the vertical coupling portion second end. An interlocking hook is disposed on the vertical coupling portion's rear surface. The interlocking hook extends through the slot in the vertical support.

(9) In another embodiment, the bracket is coupled to a corner turn assembly. The corner turn assembly has a first sidewall, a second sidewall coupled to the first sidewall, and a rear sidewall coupled to both the first sidewall and the second sidewall. Interlocking hooks are affixed to the exterior surface of the rear sidewall. The interlocking hooks extend through the slot in the vertical support. A first vertically aligned corner assembly slot is on the exterior surface of the first sidewall. The corner assembly slots accept the interlocking hooks of the bracket to connect the bracket to the corner turn assembly.

(10) The system includes a clamping block coupled to the bracket. The clamping block has a plurality of throughways that extend through the clamping block from a first face to a second face. The positive bus line extends through a first throughway and the negative bus line extends through a second throughway.

(11) The system also includes a multi-tap connector formed from a conductive material. The connector has a housing formed as an elongated hollow body with an exterior surface, a first

opening, a second opening, and a socket extending from the first opening to the second opening. The trunk bus extends through the socket from the first opening to the second opening. The connector also has a passage that extends from the exterior surface through the housing to the socket. A fastener, such as a shear bolt, is housed within the passage and frictionally engages the trunk bus within the socket.

(12) The connector also includes a plurality of taps that are each formed as a conduit that extends through the connector housing to the socket. A conductor line, such as photovoltaic wires, is housed within each of the taps. The conductor line is electrically connected to the electrical power generation unit and the trunk bus. In another embodiment, the connector is at least partially encased within an insulating material that is molded thermoplastic elastomer disposed on the connector by heat shrinking.

(13) In another embodiment, the clamping block include a first row of throughways above a second row of throughways linearly arranged along the length of the clamping block. The clamping block can be modular such that the first row of throughways is separable from the second row of throughways. The clamping block can also be made of an ultra-high molecular weight polyethylene.

(14) The positive trunk bus line and the negative trunk bus line comprise a conductive cable that is partially covered by a first insulating layer made of cross-linked polyethylene material and a second insulating layer partially covering the first insulating layer made of a high-density polyethylene material. The first insulating layer is non-colored and has a thickness between 5 mils to 15 mils and the second insulating layer has a thickness between 20 mils to 30 mils.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Features, aspects, and advantages of the present invention are better understood when the following detailed description of the invention is read with reference to the accompanying figures, in which:

(2) FIG. 1 illustrates a conventional configuration for interconnecting solar panel arrays.

(3) FIG. 2 illustrates trunk bus installation through trenching techniques.

(4) FIG. 3A illustrates a conventional configuration for supporting a trunk bus using hangers.

(5) FIG. 3B illustrates a conventional configuration for supporting a trunk bus using hangers.

(6) FIG. 4 depicts installation of a trunk bus using a bracket according to one embodiment of the disclosed bus system.

(7) FIG. 5 depicts installation of a trunk bus using a bracket according to one embodiment of the disclosed bus system.

(8) FIGS. 6A and 6B depict installation of a trunk bus using a bracket according to one embodiment of the disclosed bus system.

(9) FIG. 6C depicts a bracket according to one embodiment of the system.

(10) FIG. 7 depicts a bracket according to one embodiment of the system.

(11) FIG. 8A is a perspective side view of a trunk bus bracket according to one embodiment of the system.

(12) FIG. 8B is a perspective side view of a trunk bus bracket according to one embodiment of the system.

(13) FIG. 9 depicts a trunk bus, a bracket, and a clamping block according to an embodiment of the system.

(14) FIG. 10 depicts a modular clamping block.

(15) FIG. 11A depicts a mounted corner turn assembly and C-bracket.

(16) FIG. 11B illustrates the function of a corner turn assembly.

- (17) FIG. 12A illustrates a perspective view of a corner turn assembly.
- (18) FIG. 12B illustrates a side view of a corner turn assembly.
- (19) FIG. 12C illustrates an end view of a corner turn assembly.
- (20) FIG. 12D illustrates a rear view of a corner turn assembly.
- (21) FIG. 13 depicts an aluminum coated, steel core member extending through a bus line.
- (22) FIG. 14 is an example multi-tap connector according to one embodiment.
- (23) FIG. 15 is an example multi-tap connector according to one embodiment.
- (24) FIG. 16 is an example multi-tap connector according to one embodiment.
- (25) FIG. 17A is an example heat-shrink boot encapsulating a multi-tap connector according to one embodiment.
- (26) FIG. 17B is an example heat-shrink boot encapsulating a multi-tap connector according to one embodiment.
- (27) FIG. 18 illustrates temperature cycling test results of a connector.
- (28) FIG. 19 illustrates temperature cycling test results of a multi-tap connector according to one embodiment.
- (29) FIG. 20 illustrates temperature cycling test results of a multi-tap connector according to one embodiment.

DETAILED DESCRIPTION

- (30) The present invention will now be described more fully hereinafter with reference to the accompanying pictures in which example embodiments of the invention are shown. However, the invention may be embodied in many different forms and should not be construed as limited to the representative embodiments set forth herein. The example embodiments are provided so that this disclosure will be both thorough and complete and will fully convey the scope of the invention and enable one of ordinary skill in the art to make, use, and practice the invention.
- (31) Relative terms such as lower or bottom; upper or top; upward, outward, or downward; forward or backward; and vertical or horizontal may be used herein to describe one element's relationship to another element illustrated in the figures. It will be understood that relative terms are intended to encompass different orientations in addition to the orientation depicted in the drawings. By way of example, if a component in the drawings is turned over, elements described as being on the "bottom" of the other elements would then be oriented on "top" of the other elements. Relative terminology, such as "substantially" or "about," describe the specified materials, steps, parameters, or ranges as well as those that do not materially affect the basic and novel characteristics of the claimed inventions as whole (as would be appreciated by one of ordinary skill in the art).
- (32) Solar Panel Array Bus Systems
- (33) Examples of conventional configurations for electrically interconnecting solar panel arrays are illustrated in FIGS. 1-3. The individual solar panels **106** are arranged into panel arrays **110**. Each solar panel array **110** outputs approximately 6 A to 10 A of current through a photovoltaic ("PV") wire harness **114** that leads to a combiner box **120A & 120B**. The current from each panel array **110** is combined within the combiner box **120A & 120B** and fed to a trunk bus **130**. The trunk bus **130** carries the combined current to an inverter **140** that connects to a disconnect box and to a local power grid.
- (34) Note that the example configurations shown in FIGS. 1-3 are illustrative and not intended to be limiting, and there are other potential configurations that can be improved using the inventive systems, components, and methods described in this specification. For instance, FIG. 1 shows three panel arrays **110** and corresponding PV wire harnesses **114** feeding combiner boxes **120A & 120B**, but in some configuration there could be ten, twenty, or another predetermined number of solar panel arrays **110** connected to a combiner box **120A & 120B**. Moreover, the PV wire harnesses **114** could be connected to an initial feeder bus that extends along a row of panels arrays **110**. The feeder bus then connects the panel arrays **110** to the combiner boxes **120A & 120B** and trunk bus **130**, which is also known as a "home run" to the power station including the inverter **140**. In those

cases, both the trunk bus **130** and the initial feeder bus could be improved with the self-supporting bus **130** disclosed herein or the use of multi-tap connectors disclosed herein.

(35) The trunk bus **130** includes one or more conductor lines in combination with other lines that run in parallel and provide mechanical support and grounding protection. The trunk bus **130** can include a positive trunk bus line **134** and a negative trunk bus line **136** made from available photovoltaic cable. The positive trunk bus line **134** and negative trunk bus line **136** can be formed as electrically conductive stranded cables surrounded by one or more insulating layers. For higher amperage applications, the positive and negative trunk bus lines **134** & **136** are formed as bus bars.

(36) The positive and negative trunk bus lines **134** & **136** are optionally housed within the same or within separate conduits (i.e., a hollow plastic or metal tube that houses the bus lines) for mechanical stability and protection against weather elements. The trunk bus **130** can also include a ground cable **132** (also known as a bonding cable). Additionally, the trunk bus **130** can be formed with a grounding line that electrically connects to each of the combiner boxes to protect against instances where a grounded combiner box becomes inadvertently energized.

(37) With reference to FIG. 1, multiple combiner boxes **120A** & **120B** may be connected in series along the trunk bus **130** such that the current becomes increasing larger across each combiner box **120A** & **120B** nearing the inverter **140**. To illustrate with reference to the simplified example configuration shown in FIG. 1, the first combiner box **120A** accepts input from three solar panel arrays **110** that each have a current output of 6 A-10 A. Assuming for the sake of illustration that each panel array **110** has an output of 10 A at 600 volts direct current (“VDC”). The output of the first combiner box **120A** to the trunk bus **130** is then 30 A at 600 VDC. If a second row of three panel arrays (not shown) is connected to the second combiner box **120B**, then the output at the second combiner box **120B** is 60 A at 600 VDC. Thus, the current and power output, which is proportional to current (i.e., $\text{Power} = \text{Current} \times \text{Voltage}$), are both doubled at the output of the second combiner box **120B**.

(38) The increasing current at the output of each combiner box leads to combiner box overheating and failure and limits the number of solar panel arrays that can be connected along a trunk bus **130**. Combiner boxes are difficult to service safely because of the larger currents handled, and connection points within the combiner boxes must be torqued and otherwise installed precisely to avoid current loss that leads to overheating. Moreover, the increasing current requires larger conductor sizes for the trunk bus **130** that can become challenging and costly to install. FIG. 2 depict a conventional trenching technique for installing a trunk bus.

(39) As illustrated in FIG. 2, trenching techniques entail burying the trunk bus **130** within a trench **144** created proximal to rows of solar panel arrays **110**. Disadvantages of trenching techniques include the need for heavy equipment to form the trench **144** and the difficulty of repairing or servicing the trunk bus **130** after it has been buried.

(40) Self-Supporting Bus Systems

(41) Above-ground bus systems offer advantages that include the ability to access the trunk bus **130** for servicing and repair. A conventional above-ground system is shown in FIGS. 3A-3B that relies on thousands of hangers **150** to support the trunk bus **130** from a ground or bonding cable **132**. The hangers **150** add significant weight and cost to the system and must be installed in close proximity to one another (e.g., every 18 inches) to support the substantial lengths and weights of trunk buses **130**. A 500 MW solar field could require between 500,000 to 750,000 hangers that must be manufactured and installed at a substantial cost of time and money.

(42) Installing the hangers requires running the various trunk bus **130** lines through each individual hanger **150**, and in some cases, each hanger **150** must be bent or twisted to create a closed loop **152**, such as the example hangers shown in FIG. 3B. Given the substantial number of hangers **150** required and the significant labor involved with installing each hanger, the use of hanger installation techniques can involve substantial material and labor costs as well as installation time that can be avoided using the present self-supporting bus solutions.

(43) The present self-supporting bus solutions eliminate the material costs and labor required to install hanger systems by coupling the trunk bus **130** lines to vertical supports **115**, as shown in the attached Figures. The trunk bus **130** is coupled to a vertical support **115** using brackets (**117**, **160**, **170**, **180**) or other suitable coupling means that include, without limitation, hooks, turnbuckle tensioners, wire clamps, wire claps, or fasteners driven through the trunk bus **130**. In addition to substantial savings of installation time, labor costs, and material costs, eliminating the need for hangers has the benefit of permitting the ground cable **132** to be a pure electric ground without the need for tension to support the trunk bus **130**. The system does not, therefore, require independent dead ends or anchor points at corners or points where the trunk bus changes direction, which further reduces installation time and cost given there is no longer a need to set up the anchor points. The self-supporting solution allows for seamless installation of a single ground cable **132** from the furthest point in a solar field.

(44) Conventional brackets, such as the elbow bracket shown in FIG. 4, must be installed on a vertical support **115** using hardware that includes nuts, bolts, and washers, and the bracket utilizes a copper lug that accommodates the ground cable **132**. The trunk bus **130** is installed by seating the ground cable **132** within the clip **172** so that the ground cable **132** supports the weight of the trunk bus **130**. In another embodiment, the example bracket **117** shown in FIG. 5 includes a notch **118** that accommodates the ground cable **132**. The trunk bus **130** is installed by seating the ground cable **132** within the notch **118** so that the ground cable **132** supports the weight of the trunk bus **130**.

(45) In one example embodiment, the brackets (**117**, **160**, **170**, **180**) are mounted to a vertical support **115** installed at each row of solar panels, or installed at a distance of every 18 to 25 feet apart, while still being able to accommodate the length and weight of the trunk bus **130**. In other embodiments, the brackets (**117**, **160**, **170**, **180**) are replaced with grounding clamps that securely couple the trunk bus **130** to vertical supports **115**.

(46) One embodiment of the bracket **160** depicted in FIGS. 6A-6C includes a head **161** portion, a neck **163** portion, and a planar body portion **165** with apertures **162** disposed in the body portion **165**. The body **165** is elongated with a top surface and a bottom surface such that the apertures **162** extend through the body from the top to the bottom surface. The body **165** has a first axis (A to A') extending from the head **161** along the length of the elongated body **165** and a second axis (B to B') extending across the width of the body **165**.

(47) In some embodiments, the width of the bracket (B to B') can vary across the length of the body **165** (A to A'). For instance, the body **165** can be formed with a larger width near the head **161** and neck **163** to provide additional rigidity and strength. The portion of the bracket **160** near the head **161** is secured to the vertical support **115** and becomes load bearing when a trunk bus **130** is fixed to the bracket **160**, and the additional width mitigates against bracket failure and lateral movement across the B-B' direction.

(48) To install the bracket **160** to a vertical support **115**, the bracket is rotated about the first axis such that the apertures are perpendicular to the ground. As illustrated in FIGS. 6A-6B, the head **161** and neck **163** are inserted through a vertically aligned slot **116** in the vertical support **115**, and the bracket **160** is rotated ninety degrees to lock the bracket **160** into place. The trunk bus **130** is secured to the planar body **165** portion using a lug or a fastener, such as a bolt or screw, that is inserted through the aperture **162** and tightened to clamp down on the trunk bus **130**. In particular, the ground cable **132** can be placed across the top of the body portion **165** within a clip, such as the clip **172** shown in FIGS. 8A-8B, and the clip **172** can be clamped with a fastener to hold the ground cable **132** and the remainder of the trunk bus **130** in place.

(49) In another embodiment shown in FIG. 7, the bracket **170** is formed in approximately an "L-shape" with a planar body **175**, apertures disposed in the body **175**, a rabbeted portion **173**, and a stem **174**. To secure the bracket **170** to a vertical support **115**, the body **175** is angled upward, the stem **174** is inserted through a vertically aligned slot in the vertical support **115** until the rabbeted

portion **173** meets the slot edges, and the bracket **170** is rotated downward such that the body **175** is approximately parallel to the ground. The trunk bus **130** is fixed to the body **170** using threaded fasteners and/or a clip, such as the clip **172** and fastener shown in FIGS. **4** and **8A-8B**. In another embodiment, the bracket **170** is secured to the vertical support **115** using a fastener, such as screw or bolt, as depicted in FIG. **4**.

(50) Another embodiment uses the C-shaped bracket **180** shown in FIGS. **8A-8B**. The C-bracket **180** shown in FIGS. **8A-8B** includes: (i) a first portion **182**, also referred to herein as a ground cable support portion, extending perpendicular from the front surface of the vertical coupling portion first end that is used to support a ground cable or bonding cable between spans of vertical supports **115** (ii) a clip **172** disposed on the ground cable; (iii) a second portion **183**, also referred to herein as a trunk bus support portion, extending perpendicular from the front surface of the vertical coupling portion first end that is used to secure the clamping blocks **190** and trunk bus **130**; (iv) one or more clamping block coupling apertures **185** that accommodate fasteners that hold the clamping blocks **190** in place; (v) a third portion **181**, or vertical coupling portion, with a first end, a second end, a front surface and a rear surface, that couples to the vertical supports **115**; and (iv) interlocking hooks **186** disposed on the vertical coupling portion rear surface that extends through the vertically aligned slot **116** in the vertical support **115**.

(51) The C-shaped bracket **180** is secured to the vertical supports **115** by inserting the interlocking hooks **186** into the vertically aligned slots formed in the vertical supports **115**. Given the relatively small width of the bracket **180**, the bracket **180** can be coupled to the web or flange section of a vertical support **115**. The ground cable **132** is coupled to the first portion **182** using the clip **172**, as shown in FIG. **9**. The trunk bus support portion **183** is coupled to clamping blocks **190** using fasteners **194** that extend through clamping block coupling apertures **185**. The C-bracket **180** can also include mechanical screws that pierce the insulation of the trunk bus lines **130** to hold the trunk bus lines in place.

(52) The clamping blocks **190** include one or more rows with varying numbers of cable throughways **192** in each row. The clamping block **109** has a length that goes along the row of cable throughways **192** and has a first face and a second face. The cable throughways **192** extend through the clamping block **190** from the first face to the second face. In this way, each row is linearly arranged along the length of the clamping block **190**. The clamping block shown in FIGS. **9** and **10** include six cable throughways **192** in each row, but more or less cable throughways **192** can be used in each row. The clamping blocks **190** are modular such that one or more rows of throughways **192** can be stacked on top of one another to accommodate more trunk bus **130** lines, such as the clamping block **190** shown in FIG. **10**. The rows of throughways **192** can be secured together with dowels, pegs, detents, bolts or other fasteners that extend through each row into the clamping block material.

(53) The clamping blocks **190** provide spacing between trunk bus lines to promote heat dissipation. The clamping block **190** embodiment shown in the attached figures accommodates 20% more current without demonstrating increased temperature as compared to trunk bus lines that are integrated or bundled together in close proximity. Lower operating temperatures provides advantages that include mitigating against insulation material degradation and reduced shrink back at connection points. The cable clamping blocks **190** can be formed with an ultra-high molecular weight polyethylene with additives for enhanced ultraviolet light resistance, such as titanium oxide, carbon black, benzophenones, triazoles, organic nickel compounds, and hindered amine light stabilizers ("HALS").

(54) The brackets (**160**, **170**, **180**) shown in FIGS. **6A-6C**, **7**, and **8A-8B** can be made with galvanized high strength steel and have the advantage that the brackets (**160**, **170**, **180**) can be secured to the vertical support **115** without using threaded fasteners. Such installation is expedient and has no hardware cost, which provides substantial time savings and installation cost for systems where the brackets are installed a distance of every 18 to 25 feet. The brackets **160** & **170** can

support up to four trunk bus lines **134** & **136**. The bracket **180** combined with a stacked clamping block **190** can support up to 36 trunk bus lines, as illustrated in FIG. **10**.

(55) Eliminating hangers has the consequence that the ground cable **132** does not need to be placed in tension to be load bearing to support the trunk bus **130**. In conventional systems, the ground cable **132** is placed under tension by securing the ground cable **132** to anchor points, including where the trunk bus **130** changes direction, such as moving at a right angle. This also means that the ground cable **132** cannot be continuous but is broken into section that run between anchor points.

(56) Without the need for anchor points, the present self-supporting solution can couple the brackets (**160**, **170**, **180**) to vertical supports **115** using a corner turn assembly **200** that permits the brackets to be mounted at an angle such that the brackets do not extend straight outward from the face of the vertical support **115**. FIGS. **11A-11B** illustrates use of a corner turn assembly **200** that is coupled to a C-bracket **180** to permit the trunk bus **130** to seamlessly extend around an approximately 90-degree angle. The ground cable **132** can therefore be formed as a single continuous line without the need to secure the ground cable **132** to anchor points under tension.

(57) FIGS. **12A-12D** depict different views of the corner turn assembly **200**, which includes interlocking hooks **186** on a rear face of the assembly **200** where the hooks **186** fit into the vertically aligned slots formed in the vertical supports **115**. The corner turn assembly **200** also includes vertically aligned corner assembly slots **202** formed in the angled faces of the assembly **200** where the slots **202** accept fasteners or interlocking hooks **186** to couple the brackets to the corner turn assembly **200**.

(58) Compared to traditional methods of trunk bus installation, the trunk bus **130** disclosed in this specification takes substantially less labor and time to install and results in substantial savings in material costs by removing the need for hangers **150** and a separate ground cable **132**. More specifically, data gathered by Applicant to date shows that installation costs can be reduced by 25% to 50% using the trunk bus **130** construction and installation techniques. In addition to “home run” trunk busses that connect to a centralized power system, the trunk bus **130** and installation techniques disclosed herein can be applied to the construction of other types of electrical busses, including bus lines that connect individual panel arrays **110** or bus lines in other applications besides solar power generation.

(59) Ruggedized SolarTough® Cable

(60) The positive trunk bus line **134** and negative trunk bus line **136** can each be made with a SolarTough® ruggedized cable. The SolarTough ruggedized cable includes a conductive cable surrounded by an outer, first insulation layer and an inner, second insulation layer. The thickness of the insulation layers varies according to the size of the conductor cable that the insulation layers surround. Those of skill in the art will appreciate that the example embodiments described below are not intended to be limiting, and the SolarTough ruggedized cable can be constructed with materials and layer thicknesses that achieve comparable functionality and performance.

(61) In one example embodiment, the first insulation layer can be made from cross-linked polyethylene (“XLPE”). The first insulation layer is itself surrounded by a second insulation layer of high-density polyethylene (“HDPE”) that is also referred to as a “jacket.” The SolarTough cable is generally used for applications with voltages between about 600V and about 2000V. The sizing of the SolarTough cable varies according to electrical capacity, as illustrated below in Table 1. Sizing is provided in KCMil or “kilo circular mils,” which is a unit of area, equal to the area of a circle with a diameter of one mil (one-thousandth of an inch).

(62) TABLE-US-00001 TABLE 1 SolarTough Cable Sizing Size Conductor Insulation Jacket Overall (AWG/ # of Diameter Thickness Thickness Diameter KCMIL) Strands (Mils) (Mils) (Mils) (Mils) 8 7 132 73 13 304 6 7 166 73 13 338 4 7 209 73 13 381 2 7 264 73 13 436 1 19 300 90 16 512 1/0 19 362 90 16 574 2/0 19 410 90 16 622 3/0 19 457 90 16 669 4/0 19 496 90 16 708 250 37 543 102 18 783 300 37 569 102 18 809 350 37 620 102 18 860 400 37 657 102 18 897 500 36 744

102 18 984 600 61 812 115 20 1083 700 61 877 115 20 1148 750 60 917 115 20 1188 1000 60
1059 115 20 1330 1250 90 1196 132 23 1507

(63) The electrical properties in turn vary according to cable sizing, as shown in Table 2.

(64) TABLE-US-00002 TABLE 2 SolarTough Cable Electrical Properties Size Approx. Bend
(AWG/ Weight Resistance Resistance Ampacity Radius KCMIL) (lb/kft) (Ω /kft 20° C.) (Ω /kft 90°
C.) 90° C. (in) 8 53 1.0504 1.3467 60 2.4 6 68 0.6610 0.8474 85 2.7 4 91 0.3300 0.4231 115 3.0 2
124 0.2610 0.3346 150 3.5 1 166 0.2070 0.2654 175 4.1 1/0 201 0.1640 0.2103 205 4.6 2/0 237
0.1300 0.1667 235 5.0 3/0 280 0.1030 0.1321 270 5.4 4/0 330 0.0810 0.1038 315 5.7 250 399
0.0694 0.0890 355 6.3 300 453 0.0578 0.0741 395 6.5 350 512 0.0495 0.0635 445 6.9 400 569
0.0434 0.0556 480 7.2 500 685 0.0347 0.0445 545 7.9 600 829 0.0289 0.0371 615 8.7 700 942
0.0248 0.0318 670 9.2 750 1000 0.0231 0.0296 700 9.5 1000 1275 0.0173 0.0222 845 10.6 1250
1611 0.0138 0.0177 965 12.1

(65) The SolarTough cable is manufactured using a specially designed extrusion head that creates a curable product, which meets the UL specification **4703**. The high-density polyethylene compound for the second insulation layer is extruded simultaneously over a pure cross-linked polyethylene material for the inner, second insulation layer. This creates a jacket where both layers are fully cross-linked resulting in significantly tougher jacket with superior surface abrasion resistance as well as color fastness in the outer layer.

(66) The inner first insulation layer is black while the outer second insulation layer is colored to indicate, for example, whether the cable is a positive or negative bus line. The concentration of color particles in the outer layer delivers improved color fastness while protecting against degradation from UV light. The lack of color particles (particularly red) in the black, inner insulation layer means the phenomenon of porosity, which is normally seen with red color particles mixed throughout the insulation, is no longer created. This leads to longer life and more reliable insulation of the conductor. Those of skill in the art will appreciate that the example construction is not intended to be limiting, and other suitable thicknesses and insulators could be used that are capable of withstanding high-voltage applications.

(67) In one embodiment, the SolarTough cable replaces some of the aluminum stranded conductors inside the cable with an aluminum coated, steel core member **131** shown in FIG. **13**. The steel core member increases rigidity and strength of the bus line, which is particularly useful for applications where the bus line is suspended between vertical supports **115**. The added strength from the steel core member **131** allows the bus line to be self-supported and span up to thirty feet between vertical supports **115** without the need for hangers that are used in conventional systems to support the trunk bus line from the ground cable **132**.

(68) Notably, steel is not as conductive as aluminum, which can result in a conductivity loss by replacing some of the aluminum stands in a cable with the steel core. Coating the steel core member with aluminum mitigates against the loss of conductivity and in part takes advantage of the “skin effect” where current tends to become concentrated near the exterior surface of a cable conductor strand with a circular cross section. Thus, coating the steel core member with aluminum places a higher conductivity material where the current tends to concentrate. Utilizing an aluminum coating can recapture about 70% of the current capacity that would have otherwise been lost when using a steel core member itself without the coating.

(69) Multi-Tap Connector

(70) Once the trunk bus **130** is installed, the solar panel arrays **110** can be connected to the trunk bus **130** using multi-tap penetrating connectors **210**, such as the shear-bolt connectors shown in FIGS. **14**, **15**, **16**, and **17A-17B**. The connectors **210** are disposed about the trunk bus **130** lines and include a plurality of taps **212** for receiving portions of the PV wire harnesses **114**. The multi-tap shear bolt connectors **210** show favorable temperature performance under high current conditions, and importantly, the connectors **210** obviate the need for combiner boxes that are prone to overheating and failure.

(71) The connectors **210** include a hollow body portion (e.g., a housing **214**) with one or more socket openings **218** defining the interior of the housing, or socket. The socket is a continuous channel extending through the entire length of the hollow body portion. The socket is configured to accommodate and firmly secure a conductor (i.e., a cable) that extends through the length of the hollow body portion, such as the positive trunk bus line **134** or the negative trunk bus line **136**. The socket may define a continuous channel through the housing **214**, or the socket can be formed from halves or shells of the connector housing **214** joined to form a channel.

(72) Conductors (e.g., the positive and negative trunk bus lines **134** & **136**) are secured within the socket by shear bolt fasteners **220** that extend through passages **215** in the connector housing **214** to penetrate through the conductor insulation and into the interior of the conductor material. In this manner, the fasteners **220** establish metal-to-metal conductive contact between the conductor interior and the connector housing **214**.

(73) The head **222** of the shear-bolt fasteners **220**, and in some cases part of the stem **224**, is designed to shear off when the fasteners **220** are subjected to a predetermined torsion load. At least a portion of the shear-bolt stem **224** remains within the passage **215** after shearing. Preferably, the fastener **220** shears such that the stem **224** does not extend beyond the exterior surface of the connector housing **214**. Notably, the connector **210** is not limited to a shear bolt connector, and the taps **212** can likewise be formed in other types of connectors and used to support the bus systems disclosed herein.

(74) The solar panel arrays **110** are placed in electrical communication with the multi-tap connector **210** and trunk bus **130** by inserting portions of the wire harnesses **214** into the connector taps **212**. In particular, insulation is stripped from a portion of photovoltaic wire extending from the wire harness **214**, and the exposed photovoltaic wire is placed within the tap **212** and held in place with a set screw (not shown). This places the photovoltaic wire harness **114** in electrical communication with the connector housing **214** and the conductor within the socket (i.e., the positive or negative trunk bus line **134** & **136**).

(75) The example multi-tap shear bolt connector **210** embodiment shown in the attached Figures allows four panel arrays **110** to connect to an initial feeder bus line or a trunk bus line **130**, which increases the number of connection points on a bus line compared to use of combiner boxes. Increasing the number of connection points avoids the substantial heat generation that occurs at combiner boxes that might otherwise accept input from tens of panel arrays **110**.

(76) Once the multi-tap connector **210** is installed on a bus line, it can be sealed with an insulating material to provide mechanical support, electrical isolation, and protection from moisture (i.e., ingress protection) or other weather elements. The sealing can be accomplished prior to field installation using, for example, a molded thermoplastic elastomer, a molded thermoset elastomer, or a heat-shrink boot that encapsulates the connector **210**. An example of a heat-shrink boot encapsulating the connector **210** is depicted in FIGS. **17A-17B**. FIG. **17B** illustrates the insulation partially covering four conductors (e.g., photovoltaic wires) extending from the taps of the multi-tap connector.

(77) In other embodiments, the connector **210** can be provided with a sealing kit adapted for field use and installation. The field sealing kit can include an insulating sleeve, a tape, or a semi-solid paste or gel that is applied to the connector and heat-dried to shrink or conform the insulating material to the connector **210**. The insulating material can be a cross-linked polyethylene or another suitable insulating polymer material. The connection can also be sealed utilizing a factory-installed heat shrink polyolefin boot that is custom designed to encapsulate the connector body and the taps **212**, as shown in FIGS. **17A-17B**.

(78) Multi-tap Connector Experimental Results

(79) Temperature cycle testing was performed on the above-described the multi-tap shear bolt connector. The connector embodiment tested was sized at 1,000 kilo-circular mils ("kcmil") and designed to accommodate a current of 200 A. The taps were sized to receive 8 American Wire

Gauge (“AWG”) conductors and rated to handle current loads of approximately 40 A per tap. This sizing ensures the individual taps are well suited to receive photovoltaic wire coming from the solar panel arrays that typically carry currents between 6 A to 10 A per array under routine field operating conditions.

(80) The first test, Test 1, was conducted on a 1,000 kcmil connector with four shear bolts and a full connector with eight shear bolts. Test 1 utilized two pieces of 750 kcmil aluminum cable coupled at one end to the connector and coupled at the other end to a current source. No photovoltaic cables were inserted into taps on the connectors. Current was first applied at 400 A and increased 50 A every eight hours until reaching a maximum current of 580 A. The current was applied over the course of a week with the current turned on during the day and off at night to simulate real-world solar field current generating conditions.

(81) Thermocouples were placed at various locations on the connector and conductors to measure temperature. The temperature readings are shown in FIG. 18 where the channel readings are as follows: (1) CH2—inside the connector close to a shear bolt; (2) CH3—on the insulation of a 750 kcmil cable close to the connector; (3) on the conductive material of a 750 kcmil cable by inserting the thermocouple through a cut on the cable; (4) CH 5, CH6, and CH7 at the same locations as (1) to (3) on the full connector; and (5) CH8 as the ambient temperature. The experimental results are reflected in FIG. 18 and show a maximum temperature of 45.2 degrees Celsius ($^{\circ}\text{C}$.).

(82) A second experiment, Test 2, was conducted using a full 1,000 kcmil connector with four taps connected to 8 AWG photovoltaic wires of equal length. Current was run through the 8 AWG photovoltaic wires with no 750 kcmil conductor cable inside of the connector socket. During temperature cycling, the connector was subjected to a 240 A current during the day with the current being removed at night to simulate real-world solar power generation cycling conditions. The current was assumed to be evenly distributed among the four taps at 60 A per tap. The current was delivered through 8 AWG wire secured to each tap. The temperature curves are shown in FIG. 19 where each curve represents the temperature at a different location on the connector and conductors.

(83) Temperature was measured using thermocouples affixed to various locations on the connector and the conductors. With reference to FIG. 19, temperatures at the four taps are represented by curves CH2, CH5, CH8, and CH11. The temperature was also measured at the top of the 8 AWG wire insulation close to the connector as represented by curves CH3, CH6, CH9, and CH12. Temperature was further measured on the actual conductor material of the 8 AWG wires by inserting a thermocouple through a slit in the insulation, and conductor temperature is represented by curves CH4, CH7, CH10, and CH13. The ambient temperature is represented by curve CH14.

(84) The experimental test results showed that the multi-tap shear bolt connector was able to conduct electric current without excessive temperature rise. The highest temperature at one of the 8 AWG inputs was approximately 32.3 degrees Celsius ($^{\circ}\text{C}$.) when the ambient temperature was 22.6 $^{\circ}\text{C}$.

(85) Lastly, a third experiment, Test 3, was conducted by applying a current to the 750 kcmil aluminum photovoltaic conductor cable housed within a connector socket and the four 8 AWG photovoltaic wires secured within the connector taps. Test 3 began with 60 A applied to each of the 8 AWG photovoltaic wires and a current of 505 A applied to the 750 kcmil conductor cable. Current was increased on the 8 AWG photovoltaic wires to 70 A and increased to 590 A on the 750 kcmil conductor cable.

(86) Thermocouples were used to measure the temperature at various locations with the corresponding temperature readings shown in FIG. 20 as follows: (1) CH2—measured at a connection point of one of the 8 AWG photovoltaic wires and the connector; (2) CH3—on the conductive material of the 8 AWG photovoltaic wire by inserting a thermocouple through a cut in the wire insulation; (3) CH4—on top of the 750 kcmil conductor cable insulation close to the connector; (4) CH5—on the surface of the connector near the shear bolts; (5) CH6—on the

conductive material of the positive current 750 kcmil cable by inserting a thermocouple through a cut in the wire insulation; (6) CH7—on the insulation of the positive side of the 750 kcmil cable near location CH6; (7) CH8—on the conductive material of the negative side of the 750 kcmil cable by inserting a thermocouple through a cut in the wire insulation; (8) CH9—on the insulation of the negative side of the 750 kcmil cable near location CH8; and (9) CH10—the ambient temperature.

(87) FIG. 20 depicts the experimental results for the high current conditions of Test 3. The highest temperature was 72.8° C. at the 750 kcmil conductor cable when the ambient temperature was 27.4° C. The 8 AWG photovoltaic wires showed stable temperature performance and stayed below 30° C. All measured temperatures stayed below 90° C. indicating that the connector could maintain performance without heat failure under high current conditions.

(88) Although the foregoing description provides embodiments of the invention by way of example, it is envisioned that other embodiments may perform similar functions and/or achieve similar results. Any and all such equivalent embodiments and examples are within the scope of the present invention.

Claims

1. An electrical bus system comprising: (a) a power generation unit; (b) a trunk bus that comprises (i) a positive bus line, (ii) a negative bus line, and (iii) a first aluminum coated steel core member running through the interior of the positive bus line and a second aluminum coated steel core member running through the interior of the negative bus line; (c) a bracket coupled to a vertical support, wherein the bracket comprises (i) a vertical coupling portion with a first end, a second end, a front surface, and a rear surface, (ii) a first portion extending perpendicular from the front surface of the vertical coupling portion first end, (iii) a second portion extending perpendicular from the front surface of the vertical coupling portion second end, and (iv) an interlocking hook disposed on the vertical coupling portion rear surface, wherein the interlocking hook extends through a slot in the vertical support; (d) a clamping block coupled to the bracket, wherein (i) the clamping block comprises a plurality of throughways that extend through the clamping block from a first face to a second face, and (ii) the positive bus line extends through a first throughway and the negative bus line extends through a second throughway; (e) a multi-tap connector formed from a conductive material, wherein the multi-tap connector comprises (i) a housing formed as an elongated hollow body having an exterior surface, a first opening, a second opening, and a socket extending from the first opening to the second opening, wherein the trunk bus extends through the socket from the first opening to the second opening, (ii) a passage extending from the exterior surface through to the socket, (iii) a fastener housed within the passage, wherein the fastener frictionally engages the trunk bus within the socket, (iv) a plurality of taps that are each formed as a conduit extending through the housing to the socket, and (v) a conductor line housed within each of the plurality of taps, wherein the conductor line is electrically connected to the electrical power generation unit and the trunk bus.

2. The electrical bus system of claim 1, wherein: (a) the power generation unit is a solar panel array; and (b) the conductor lines are photovoltaic wires.

3. The electrical bus system of claim 1, wherein the clamping block comprises (a) a first row of a plurality of throughways linearly arranged along a length of the clamping block; and (b) a second row of a plurality of throughways linearly arranged along the length of the clamping block, wherein the second row is above the first row.

4. The electrical bus system of claim 3, wherein the clamping block is modular such that the first row is separable from the second row.

5. The electrical bus system of claim 1, wherein the clamping block is comprised of an ultra-high molecular weight polyethylene.

6. The electrical bus system of claim 1, wherein: (a) the bracket comprises a head, a neck, a planar body, and an aperture; (b) the head and the neck extend through a slot in the vertical support; and (c) the clamping block is secured to the planar body of the bracket.
7. The electrical bus system of claim 1, wherein: (a) the bracket comprises a planar body with a free end and a coupling end, a rabbeted portion extending from the coupling end, and a stem extending from the rabbeted portion opposite the planar body; and (b) the stem extends through a slot in the vertical support.
8. The electrical bus system of claim 1, wherein the bracket is coupled to a corner turn assembly comprising: (a) a first sidewall, a second sidewall coupled to the first sidewall, and a rear sidewall coupled to both the first sidewall and the second sidewall; (b) interlocking hooks disposed on an exterior surface of the rear sidewall, wherein the interlocking hooks extend through a slot in the vertical support; and (c) a first vertically aligned corner assembly slot disposed on an exterior surface of the first sidewall, wherein the corner assembly slots accepts interlocking hooks coupled to the bracket to couple the bracket to the corner turn assembly.
9. The electrical bus system of claim 1, wherein the fastener is a shear bolt.
10. The electrical bus system of claim 1, wherein: (a) the connector is at least partially encased within an insulating material; and (b) the insulating material is a molded thermoplastic elastomer disposed on the connector by heat shrinking.
11. The electrical bus system of claim 1, wherein the positive trunk bus line and the negative trunk bus line comprise: (a) a conductive cable at least partially covered by a first insulating layer, wherein the first insulating layer comprises a cross-linked polyethylene material; and (b) a second insulating layer at least partially covering the first insulating layer, wherein the second insulating layer comprises a high-density polyethylene material.
12. The electrical bus system of claim 11, wherein: (a) the first insulating layer is non-colored and has a thickness between 5 mils to 15 mils; and (b) the second insulating layer has a thickness between 20 mils to 30 mils.
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